INDEX

LOOKUP FUNCTION

[Find Specific Information in a Table or Range]

SYNTAX:	=LOOKUP (lookup_value,array)
	=LOOKUP (lookup_value, lookup_vector,[result_vector])

Problem Statement:

You are managing a small institute and have a list of employees with their respective Employee IDs, salaries, designation, and DOJ. And your manager asks about the salary of a specific employee, this is when you can use the LOOKUP function in Excel to quickly find the information.

Sales Table	
Emp ID	Salary
203	30000
201	40000
202	35000

Employee Table			
Emp ID	Emp Name	Designation	DOJ
202	Abhay	Manager	10/Jan/24
203	Rohan	Developer	23/Oct/24
201	Harsh	Designer	8/Aug/24

- What is the salary of the employee with Employee ID 203?
- What is the designation of the employee with Employee ID 202?
- What is the DOJ of the employee with Employee ID 203?

Solution:

SYNTAX USED	=LOOKUP (lookup_value,array)

Result:

Single dimensional	
Emp ID	Salary
203	30000
LOOKUP(B29,B20:C23)	

Multiple Dimensional	
Emp ID	Designation
202	Manager
LOOKUP(F29,F20:H23)	

Emp ID	DOJ
203	23/Oct/24
LOOKUP(I29,F20:I23)	

SYNTAX USED	=LOOKUP (lookup_value, lookup_vector,[result_vector])

Result:

Emp ID	Salary
203	30000
LOOKUP(B38,B20:B23,C20:C23)	

Emp ID	Designation
203	Developer
LOOKUP(F40,F19:F22,H19:H22)	

CHOOSE Function

[Select a value from a list of values based on an index number]

Syntax	= CHOOSE (index_num, value1, [value2], [value3], ...)

Problem Statement:

You advise students on which course to take based on their interests. You want to display the course name based on a student's choice.

ID	Fees	Course Name
1	2000	Excel
2	3000	Power Query
3	5000	Power BI
4	4500	Tableau
5	3500	SQL Server

- If a student wants to know about the course for option 3, what is the course name and its fees?
- Gives function Number to basic function like Sum, Average, and Max and calculates it?

Solution:

Example	=CHOOSE (N4, "Excel", "Power Query", "Power BI", "Tableau", "SQL Server")

Result:

CHOOSE Function with Sum, Average, and Max

Function No.	Fees
1	18000
2	3600
3	5000

CHOOSE(N15, SUM($P$5:$P$9), AVERAGE($P$5:$P$9), MAX($P$5:$P$9))

MATCH Function

[Find the position (row or column number) of a specific item in a range]

SYNTAX	MATCH (lookup_value, lookup_array, [match_type])

Problem Statement:

You are helping students to find the position of a specific course in a list of available courses. By using the MATCH function, you can easily determine the course's position in the list.

ID	Fees	Course Name
1	2000	Excel
2	3000	Power Query
3	5000	Power BI
4	4500	Tableau
5	3500	SQL Server

- If you want to find out the position of the course 'Tableau' in the course list, what is its position?
- If you want to find out the column number of the Fees column, how would you do it?

Solution:

Course Name	Text Position
Tableau	4

`MATCH(AK28,AM20:AM24,0)`

Fees
2

`MATCH(AK33,AK19:AM19,0)`

CHOOSE with MATCH Function

[Dynamically select values based on criteria]

SYNTAX	CHOOSE(index_num, value1, [value2], [value3], ...) MATCH(lookup_value, lookup_array, [match_type])

Problem Statement:

You are an academic advisor helping students choose courses based on their interests. You want to display the course Discount% dynamically based on the course they select from a list.

ID	Course Name	Discount %
1	Excel	10
2	Power Query	15
3	Power BI	5
4	Tableau	20
5	SQL Server	25

- A student wants to know the discount% for the course 'Tableau.' How can we find that using the CHOOSE and MATCH functions?

Solution:

Course	Discount %
Tableau	20

```
CHOOSE(MATCH(AR27,AS19:AS23,0),AT19,AT20,AT21,AT22,AT23)
```

VLOOKUP (Vertical Lookup) function

[Search for a value in the first column of a range (or table) and return a value in the same row from a specified column]

SYNTAX	Vlookup(lookup_value, table_array, col_index_num,[range_lookup])

Problem Statement:

You are the ADMIN at an institute, and you need to provide information about new employees to your team. You want to identify which employee joined most recently and gather details about their designation and joining date.

Employee Table			
Emp ID	Emp Name	Designation	DOJ
202	Abhay	Manager	10/Jan/24
203	Rohan	Sr. Manager	23/Oct/24
201	Harsh	Executive	8/Aug/24

- What is the designation of employee id 201?
- What are the designations of given employee IDs?

Solution:

Emp ID	Emp Name
201	Harsh

VLOOKUP(AE13,AE5:AH8,2,0)

Emp ID	Designation
202	Manager
203	Sr. Manager
201	Executive

VLOOKUP(AE17,$AE$5:$AH$8,3,0)

In this function, we need to use absolute cell references for the table array. This way, we can keep the table array fixed, so it doesn't change or move when we copy the formula to other cells or drag downside.

VLOOKUP with MATCH Function

SYNTAX	VLOOKUP (lookup_value, table_array, col_index_num, [range_lookup]) MATCH (lookup_value, lookup_array, [match_type])

Problem Statement:

You are a teacher at a training institute, and you have the following dataset of students and their courses

Employee Table			
Emp ID	Emp Name	Designation	DOJ
202	Abhay	Manager	10/Jan/24
203	Rohan	Developer	23/Oct/24
201	Harsh	Designer	8/Aug/24

- Use the VLOOKUP function to find out the designation for the given Emp ID
- Find out the position of Emp ID in Emp ID column
- Use a combination of VLOOKUP & MATCH to find out the Emp Name of Emp ID 201

Solution:

Emp ID	Designation
202	Manager

```
VLOOKUP(BP29,BP21:BR24,3,0)
```

Emp ID	Position
202	2

```
MATCH(BP34,BP21:BP24,0)
```

Emp ID	Emp Name
201	Harsh

```
VLOOKUP(BP40,BP21:BS24,MATCH(BQ39,BP21:BS21,0),0)
```

VLOOKUP with CHOOSE function

SYNTAX:	Vlookup(lookup_value, table_array, col_index_num,[range_lookup]) CHOOSE(index_num, value1,[value2],[value3]....)

Problem Statement:

Imagine you are managing employee records on an Excel sheet.

You have the following data on employee names, their unique IDs, and their job designations:

Employee Table			
Emp Name	Emp ID	Designation	DOJ
Abhay	202	Manager	10/Jan/24
Rohan	203	Developer	23/Oct/24
Harsh	201	Designer	8/Aug/24

Your boss asks you to quickly retrieve details for employees based on their IDs. However, you realize that Emp ID is in the second column, making it difficult to use a simple VLOOKUP (which requires the lookup column to be first).

Solution:

Emp ID	Emp Name
203	Rohan

```
VLOOKUP(BX26,CHOOSE({1,2},BY20:BY22,BX20:BX22),2,0)
```

VLOOKUP with WILDCARD Function

SYNTAX	VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])

There are two main wildcards in Excel:

- **Asterisk (*):** Represents any number of characters.
- **Question mark (?):** Represents a single character.

Problem Statement:

Course Details		
Course ID	Course Name	Discount %
PLTL4	Tableau	20
PLTL19	Adv SQL Server	10
PLTL2	Power Query	15
PLTL3	Power BI	5
PLTL6	Adv Excel	12
PLTL18	Adv Power BI	13
PLTL1	Excel	10
PLTL17	Adv Power Query	11
PLTL5	SQL Server	25

- You want to look up the discount% of any course starting with 'Adv' and require the first match.
- Your manager asks you to quickly find the discount% of any course that contains 'BI' and require the first match.
- Your manager asks you to quickly find the course of any course ID that contains 'PLTL1' and require the first match.

Solution:

*** (Asterisk):** Matches any number of characters (e.g., *abc can match '1abc', 'xyzabc', 'abc').

Discount %
10

`VLOOKUP("Adv*",CG18:CH27,2,0)`

Discount %
5

`VLOOKUP("*BI",CG22:CH30,2,0)`

? (Question mark): Matches exactly one character (e.g., abc? can match 'abcd', 'abce').

Course
Adv SQL Server

```
VLOOKUP("PLTL1?",CF24:CG33,2,0)
```

VLOOKUP with SUMPRODUCT Function

[Calculate values based on lookup results from multiple tables.]

SYNTAX	VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup]) SUMPRODUCT(array1, [array2], [array3], ...)

Problem Statement:

- You have two tables of data: one with course fees and another with course discount percentages.
- Your manager wants to see the total discount amount.

Course ID	Fees
PLTL1	2500
PLTL2	3000
PLTL3	5000
PLTL4	4000
PLTL5	3500

Course ID	Course Name	Discount %
PLTL1	Excel	10
PLTL2	Power Query	15
PLTL3	Power BI	5
PLTL4	Tableau	20
PLTL5	SQL Server	25

Solution:

Total Discount Amount
262500

```
SUMPRODUCT(CP15:CP19,VLOOKUP(CO15:CO19,CR15:CT19,3,0))
```

XLOOKUP Function

SYNTAX	XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found], [match_mode], [search_mode])

XLOOKUP is a powerful and flexible lookup function in Excel that's designed to overcome many limitations of VLOOKUP and HLOOKUP.

Introduced in Excel 365 and Excel 2019, XLOOKUP can search for a value in a range and return a corresponding result from another range.

Unlike VLOOKUP, XLOOKUP does not require the lookup column to be on the left, and it can search from the bottom up, perform exact or approximate matches, and even return multiple values.

	Parameter	Requirement	Description	Example
1	lookup_value	Required	This is the value you are searching for	Example: If you want to find a specific student's grade, the student's name would be the lookup_value
2	lookup_array	Required	This is the range or array where Excel will search for the lookup_value.	Example: If you have a list of student names in cells A2 to A10, this range is the lookup_array
3	return_array	Required	This is the range or array from which XLOOKUP will return the result if it finds a match.	Example: If the grades are in cells B2 to B10, you'd set return_array as B2 to get the corresponding grade for the matched student name
4	[if_not_found]	Optional	Specifies what to return if no match is found.	Example: Manually enter 'Not Found' to return if no match is found
5	[match_mode]	Optional	Determines how XLOOKUP matches the lookup_value. Options are: 0 (default): Exact match only. -1: Exact match or next smaller item. 1: Exact match or next larger item. 2: Wildcard match (e.g., using * or ? to represent partial matches)	

6	[search_mode]	Optional	Specifies the direction for searching. Options are: 1 (default): Search from the first to the last item. -1: Search from the last to the first item. 2: Perform a binary search in ascending order. -2: Perform a binary search in descending order.	

Problem Statement:

Emp ID	Name	Salary
101	Aditi	50000
102	Ravi	55000
103	Mohit	48000
105	Sunita	60000
110	Pooja	52000

Solution:

- **Exact Match only (match_mode=0)**

Emp ID	Salary
104	Not Found

If we want to find out salary of Emp ID 104

If this Emp ID 104 is not available in Emp ID then

```
XLOOKUP(CB50,CB40:CB44,CD40:CD44,"Not Found",0)
```

- **Exact Match or Next Smaller Item (match_mode=-1)**

Emp ID	Salary
104	48000

```
XLOOKUP(CB58,CB40:CB44,CD40:CD44,"Not Found",-1)
```

If you want the exact salary of "Emp ID 104". However, if "104" isn't found, you are okay with the salary of the next alphabetically smaller emp id

- **Exact Match or Next Larger Item (match_mode=1)**

Emp ID	Salary
104	60000

```
XLOOKUP(CB66,CB40:CB44,CD40:CD44,"Not Found",1)
```

You want to exact salary of "Emp ID 104", but if it's unavailable, you will accept the salary of the next alphabetically larger emp id

- **Wildcard Match (match_mode=2)**

Name	Salary
ra	55000

You want to find the price of any name of employee that contains"ra" in its name using a wildcard search. This will match names like: "Ravi"

```
XLOOKUP("ra*",CC40:CC44,CD40:CD44,"Not Found",2)
```

Match_mode	Scenario	Result
0	Exact match only	Not Found
-1	Exact match or next smaller item	52000
1	Exact match or next larger item	55000
2	Wildcard match	55000

Course Details

Course ID	Course Name	Discount %
PLTL4	Tableau	20
PLTL19	Adv SQL Server	10
PLTL2	Power Query	15
PLTL3	Power BI	25
PLTL6	Adv Excel	12
PLTL18	Adv Power BI	13
PLTL1	Excel	10
PLTL17	Adv Power Query	11
PLTL5	SQL Server	25
PLTL20	Excel	8
PLTL21	Excel	5

Course ID	Course Name	Discount %
PLTL6	Adv Excel	12
PLTL18	Adv Power BI	13
PLTL17	Adv Power Query	11
PLTL19	Adv SQL Server	10
PLTL21	Excel	5
PLTL20	Excel	8
PLTL1	Excel	10
PLTL3	Power BI	25
PLTL2	Power Query	15
PLTL5	SQL Server	25
PLTL4	Tableau	20

Course ID	Course Name	Discount %
PLTL4	Tableau	20
PLTL5	SQL Server	25
PLTL2	Power Query	15
PLTL3	Power BI	25
PLTL1	Excel	10
PLTL20	Excel	8
PLTL21	Excel	5
PLTL19	Adv SQL Server	10
PLTL17	Adv Power Query	11
PLTL18	Adv Power BI	13
PLTL6	Adv Excel	12

[search_mode]

1 (Default): Search from first to Last

This is the default setting. Excel searches the lookup_array starting from the first element and moving to the last useful for most situations where you want to find the first occurrence of the lookup_value

Emp ID	Name
Excel	10

```
XLOOKUP(CB89,CC105:CC115,CD105:CD115,"Not Fount",0,1)
```

-1: Search from Last to First

Excel searches the lookup_array starting from the last element and moving to the first useful when you want to find the last occurrence of the lookup_value in the list, if the lookup_array contains multiple instances of the value, this mode will return the price of the last Course name found

Emp ID	Name
Excel	5

```
XLOOKUP(CB100,CC105:CC115,CD105:CD115,"Not Fount",0,-1)
```

2 : Binary Search (Descending Order)

Performs a binary search on the lookup_array, assuming it is sorted in descending order.Similar to option 2, but for datasets sorted in descending order.

Course Name	Discount %
Excel	5

```
XLOOKUP(CB134,CK105:CK115,CL105:CL115,"No",0,-2)
```

HLOOKUP Function

[Search for a value in the top row of a table or range and returns a value in the same column from a specified row in that table or range. It is particularly useful for horizontal data lookups, where the data is organized in rows rather than columns.]

SYNTAX	HLOOKUP(lookup_value, table_array, row_index_num, [range_lookup])

Problem Statement:

You are an admin at an institute, and your manager wants to quickly find the sales of specific quarters based on course names.

Courses Sold				
Courses	Q1_Sales	Q2_Sales	Q3_Sales	Q4_Sales
Excel	58	93	96	57
Power Query	20	46	44	34
Power BI	34	59	52	35
Tableau	31	96	42	94
SQL Server	39	78	52	97

Courses	Excel	Power Query	Power BI	Tableau	SQL Server
Total Sales	304	144	180	263	266

- Find out Excel Course Qty

Solution:

Qtr.	Power BI
Q3_Sales	52

```
HLOOKUP(CP32,CP12:CT17,4,0)
```

Course	Total Sales
Power BI	180

```
HLOOKUP(CP37,CP32:CU33,2,0)
```

HLOOKUP with MATCH Function

[The HLOOKUP function combined with the MATCH function allows for dynamic row selection in horizontal lookups. MATCH helps to find the row number, making HLOOKUP flexible for changing data sets.]

SYNTAX	HLOOKUP(lookup_value, table_array, row_index_num, [range_lookup]) MATCH(lookup_value, lookup_array, [match_type])

MATCH FUNCTION:

[Find out row and column number]

Problem Statement:

If we have course sold and course fees data:

Courses Sold				
Courses	January	February	March	April
Excel	58	93	96	57
Power Query	20	46	44	34
Power BI	34	59	52	35
Tableau	31	96	42	94
SQL Server	39	78	52	97

- You want to first create course sold data in transpose
- You want to calculate the total revenue by using both datasets

Solution:

Courses	Excel	Power Query	Power BI	Tableau	SQL Server
Q1_Sales	58	20	34	31	39
Q2_Sales	93	46	59	96	78
Q3_Sales	96	44	52	42	52
Q4_Sales	57	34	35	94	97

```
HLOOKUP($CZ20,$CQ$12:$CT$17,MATCH(DA$19,$CP$12:$CP$17,0))
```

HLOOKUP with VLOOKUP Function

[Combining HLOOKUP and VLOOKUP functions allows you to dynamically retrieve data from two datasets. This is useful for calculations that rely on different lookup directions (horizontal and vertical)].

SYNTAX	HLOOKUP(lookup_value, table_array, row_index_num, [range_lookup]) VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])

Problem Statement:

Using both functions, you need to calculate the total revenue from two datasets.

Courses Sold				
Courses	Q1_Sales	Q2_Sales	Q3_Sales	Q4_Sales
Excel	58	93	96	57
Power Query	20	46	44	34
Power BI	34	59	52	35
Tableau	31	96	42	94
SQL Server	39	78	52	97

Course Fees		
ID	Course Name	Fees
1	Excel	2000
2	Power Query	3000
3	Power BI	5000
4	Tableau	4500
5	SQL Server	3500

Solution:

Calcualte the total amount				
Courses	Q1_Sales	Q2_Sales	Q3_Sales	Q4_Sales
Excel	116000	186000	192000	114000
Power Query	60000	138000	132000	102000
Power BI	170000	295000	260000	175000
Tableau	139500	432000	189000	423000
SQL Server	136500	273000	182000	339500

```
HLOOKUP(DL$24,$DK$12:$DO$17,MATCH($DK25,$DK$12:$DK$17,0),0)*VLOOKUP($DK25,$DR$12:$DS$17,2,0)
```

OFFSET with MATCH function

SYNTAX	OFFSET(reference, rows, cols, [height], [width]) MATCH(lookup_value,lookup_array,[match_type])

reference	required	explanation
rows	required	We can use here MATCH function
cols	required	We can use here MATCH function
height	optional	The number of rows you want the final range to cover. If omitted the height will match the reference
width	optional	The number of columns you want the final range to cover. If omitted, the width will match the reference

Problem Statement:

- You are maintaining Institutes expenses data, You manager want to know office rent of Apr month
- You manager want to see Laptop Rent of Qtr 2

Monthly Expenses												
Expenses	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Office Rent	3838	4125	5591	7954	4817	4125	4397	5120	5799	6800	5872	6945
Electricity	1432	1361	1107	1126	1078	1151	1453	1152	1355	1025	1138	1170
Internet	1500	1500	1500	1500	1500	1500	1500	1500	1500	1500	1500	1500
Laptop Rent	4535	4988	4212	4342	4508	4909	4046	4146	4095	4326	4648	4027
Materials	244	189	407	157	452	375	295	240	166	118	197	205
Certification	100	100	100	100	100	100	100	100	100	100	100	100
Salary	50000	50000	50000	50000	50000	50000	50000	50000	50000	50000	50000	50000
Stationery	381	455	131	485	473	354	239	147	295	187	288	436

Solution:

Expenses	Apr
Office Rent	7954

```
OFFSET(DS31,MATCH("Rent",DS32:DS36,0),MATCH("Q2_Sales",DT31:DW31,0))
```

Expenses	Qtr 2		
Laptop Rent	4046	4146	4095

```
OFFSET(DS16,MATCH(DS34,DS17:DS24,0),7,1,3)
```

Expenses	Dec
Materials	205
Certification	100
Salary	50000
Stationery	436

```
OFFSET(DT17,MATCH(DT41,DT18:DT25,0),MATCH(DU40,DU17:EF17,0),4,1)
```

OFFSET with AVERAGE, SUM, MATCH function

SYNTAX	OFFSET(reference, rows, cols, [height], [width]) AVERAGE(number1,[number2],..) SUM(number1,[number2],..) MATCH(lookup_value,lookup_array,[match_type])

Problem Statement:

- You have maintaining student admission data
- Your manager wants to know the average student enrollment for the Power BI course in Q2.
- Your manager wants to know the total student enrollment for the Excel course in Q3

Admission Data					
Month	Excel	Power Query	Power BI	Tableau	SQL Server
Jan	15	19	13	14	11
Feb	7	13	8	13	3
Mar	16	9	10	3	6
Apr	8	3	5	11	11
May	17	9	8	10	4
Jun	3	15	9	6	9
Jul	17	6	20	10	18
Aug	6	7	20	5	4
Sep	10	19	3	14	18
Oct	16	8	16	7	18
Nov	19	12	13	3	3
Dec	7	20	12	4	16

Solution:

Course	Q2
Power BI	14.33333333

```
AVERAGE(OFFSET(EH17,7,MATCH(EH35,EI17:EM17,0),3,1))
```

Month	Total Admission
Jan	72

```
SUM(OFFSET(EH17,MATCH(EH41,EH18:EH29,0),1,1,5))
```

Month	Total Admission
Jun	50

```
SUM(OFFSET(EH17,MATCH(EH46,EH18:EH29,0),2,2,2))
```

Q3	Total Admission
Oct	174

```
SUM(OFFSET(EH17,MATCH(EH52,EH18:EH29,0),1,3,5))
```

VLOOKUP, OFFSET with MATCH function

SYNTAX	OFFSET (reference, rows, cols, [height], [width])
	VLOOKUP (lookup_value, table_array, col_index_num,[range_lookup])
	MATCH (lookup_value,lookup_array,[match_type])

Problem Statement:

- You are working as an accountant in a coaching institute.
- Your CA wants to quickly know the Jan months Excel & Power BI course total GST Amt.
- Your CA wants to quickly know the Mar months Power Query & SQL Server course total GST Amt.

GST Data					
Month	Excel	Power Query	Power BI	Tableau	SQL Server
Jan	4401	6100	1965	1962	3080
Feb	5757	1698	9156	2016	2055
Mar	4437	1403	6150	1611	5166

Solution:

- **Find out total GST amount of course Excel & Power BI for Jan Months.**

Ans: 6366

Function: VLOOKUP ("Jan", ER15:EW18, MATCH ("Excel", ER15:EW15,0),0) +OFFSET (ER15, MATCH ("Jan", ER16:ER18,0), MATCH ("Power BI", ES15:EW15,0),1,1)

- **Find out total GST amount of course Power Query & SQL Server for Mar Months**.

Ans: 6569

Function: VLOOKUP ("Mar", ER15:EW18, MATCH ("Power Query", ER15:EW15,0),0) +OFFSET (ER15, MATCH ("Mar", ER16:ER18,0), MATCH ("SQL Server", ES15:EW15,0),1,1)

MAX or MIN function

SYNTAX	MAX (number1, [number2]..) MIN (number1, [number2]..)

Problem Statement:

- You have year 2023 student's total enrollment data
- You manager want to know quickly highest enrollment
- You manager want to know quickly lowest enrollment

Year 2023 Enrollment		
Month	Courses	Total Enrollment
Jan	Excel	635
Feb	Power Query	112
Mar	Power BI	318
Apr	Tableau	384
May	SQL Server	392
Jun	Excel	742
Jul	Power Query	338
Aug	Power BI	667
Sep	Excel	194
Oct	Power Query	158
Nov	Power BI	527
Dec	Excel	871

Solution:

Highest Student Enrollment
871

`MAX(ET13:ET24)`

Lowest Student Enrollment
112

`MIN(ET14:ET25)`

MAX with IF function

SYNTAX	MAX(number1,[number2]..) IF(logical_test,[value_if_true],[value_if_false])

Problem Statement:

- You have year 2023 students total enrollment data
- You manager want to know quickly highest enrollment for Excel course
- You manager want to know quickly lowest enrollment for Power BI Course

Year 2023 Enrollment		
Month	Courses	Enrollment
Jan	Excel	635
Feb	Power Query	112
Mar	Power BI	318
Apr	Tableau	384
May	SQL Server	392
Jun	Excel	742
Jul	Power Query	338
Aug	Power BI	667
Sep	Excel	194
Oct	Power Query	158
Nov	Power BI	527
Dec	Excel	871

Solution:

Course	Enrollment
Excel	871

```
MAX(IF(FA12:FA23=EZ28,FB12:FB23,""))
```

Course	Enrollment
Power BI	318

```
MIN(IF(FA13:FA24=EZ36,FB13:FB24,""))
```

MAXIFS function

[Find the maximum value based on one or more conditions. This is useful for datasets where you want to find the maximum in a filtered data]

SYNTAX:	MAXIFS(max_range, criteria_range1, criteria1, [criteria_range2, criteria2], ...)

Problem Statement:

Suppose you want to find the maximum student enrollment for a specific course like Excel. Your manager want to know the lowest enrollment for Power Query course

Year 2023 Enrollment		
Month	Courses	Course Sell
Jan	Excel	635
Feb	Power Query	112
Mar	Power BI	318
Apr	Tableau	384
May	SQL Server	392
Jun	Excel	742
Jul	Power Query	338
Aug	Power BI	667
Sep	Excel	194
Oct	Power Query	158
Nov	Power BI	527
Dec	Excel	871

Solution:

Course	Enrollment
Excel	871

```
MAXIFS(FH15:FH26,FG15:FG26,FF30)
```

Course	Enrollment
Power Query	112

```
MINIFS(FH13:FH24,FG13:FG24,FF34)
```

IF with PRODUCT function

SYNTAX	IF(logical_test,[value_if_true],[value_if_false]) PRODUCT(number1,[number2]..)

Problem Statement:

You're working on a Year 2023 student enrollment data and you need to calculate a "weighted revenue" based on course, enrollment, fees, and discount rates. You want to find the revenue for only those items with a specific month and apply a discount to get a result.

Year 2024 Enrollment				
Month	Courses	Enrollment	Per Course Fees	Discount Rate
Jan	Excel	63	2000	0.09
Feb	Power Query	12	1500	0.13
Mar	Power BI	31	3500	0.09
Apr	Tableau	38	3000	0.19
May	SQL Server	39	2500	0.20
Jun	Excel	74	2000	0.20
Jul	Power Query	33	1500	0.05
Aug	Power BI	66	3500	0.15
Sep	Excel	19	2000	0.18
Oct	Power Query	15	1500	0.13
Nov	Power BI	24	3500	0.09
Dec	Excel	18	200	0.09

- Calculate the total revenue for course after applying the discount rate, using a formula that combines PRODUCT with other functions

Solution:

Course	Total Revenue
Excel	267496.00

```
SUMPRODUCT((FQ19:FQ30=FP35)*FR19:FR30*FS19:FS30*(1-FT19:FT30))
```

Course	Total Revenue
Excel	267496.00

```
SUMPRODUCT(FILTER(FR19:FR30*FS19:FS30*(1-FT19:FT30),FQ19:FQ30="Excel"))
```

INDEX with MATCH function

SYNTAX:	INDEX(array,row_num,[column_num]) MATCH(lookup_value,lookup_array,[match_type])

Problem Statement:

Suppose you work in a company's HR department, and you have a dataset listing employees' details.

You want to retrieve an employee's **designation** and **date of joining (DOJ)** based on their **Emp Name** and **Emp ID**.

Employee Table			
Emp ID	Emp Name	Designation	DOJ
202	Abhay	Manager	10/Jan/24
203	Rohan	Developer	23/Oct/24
201	Harsh	Designer	8/Aug/24
204	Abhay	Analyst	15-Mar-24

- Retrieve the Designation and DOJ for "Abhay" with Emp ID 204.
- Retrieve the Emp ID for Employee Rohan.

Solution:

Emp ID	Emp Name	Designation
204	Abhay	Analyst

```
INDEX(FZ18:FZ21,MATCH(1,(FY18:FY21=FY26)*(FX18:FX21=FX26),0))
```

Emp ID	Emp Name	DOJ
204	Abhay	15/Mar/24

```
INDEX(GA18:GA21,MATCH(1,(FY18:FY21=FY32)*(FX18:FX21=FX32),0))
```

Name	Emp ID
Rohan	203

```
INDEX(FX17:GA21,MATCH(FX38,FY17:FY21,0),MATCH(FY37,FX17:GA17,0))
```